# Heat is not temperature

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Distinguish temperature from energy transferred because of a temperature difference.

**SUCCESS CRITERIA - I can...**

1. Compare temperature with thermal energy.
2. Identify the direction of spontaneous heat transfer.
3. Explain why amount and material matter.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “heat” and “temperature” to compare your model or result. Sentence starters are optional.

**heat**: energy transferred because of a temperature difference

**temperature**: a measure of thermal state; it is not an energy amount

**thermal energy**: internal energy associated with microscopic motion/interactions

## 01 Warm-up: bring back an earlier idea

A simplified model requires 15 J to escape, but an object has only 12 J and no later energy input. Does the model predict escape?
**Explain**.

**Sentence starter:** I predict … . My model supports this because … .

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $Q = mc\,\Delta T$.
For heating without a phase change: Q is transferred heat, m is mass, c is specific heat. Heat and temperature are different quantities.

**Worked reasoning:** A cup and a bathtub can both be at 40°C. Equal temperature does not imply equal internal energy: amount and type of material also matter.

**Your turn: Draw** equal temperature indicators but different amounts of water.
**Predict** heat-flow direction if the water samples are brought into thermal contact at the same temperature.

**Sentence starter:** Both … . They differ in … because … .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 03 Read and interpret the model

A cup and a bathtub contain the same substance at 40°C. Are their temperatures equal? Must their total internal energies be equal?

**Sentence starter:** I notice … . This supports … because … .



Cup: 40°C



Tub: 40°C

Each outline represents the labeled sample or container.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A metal spoon at 60°C is placed in water at 20°C.
**Label** both samples and draw the initial energy-transfer arrow.
**Predict** which temperature rises and which falls.

**Sentence starter:** I predict … . My model supports this because … .



Spoon: 60°C



Water: 20°C

**Marzano check - circle:** ① ② ③

## 05 Explain your reasoning

A student says “cold flows into a hot drink.” Rewrite the explanation in terms of energy transfer.

**Sentence starter:** My design shows … . It meets the requirement by … .

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A wheel rolls down a ramp, losing 24 J gravitational energy. Its center-of-mass kinetic energy gain is 20 J with negligible losses. What energy term is missing?
**Show** the energy changes and transfers; label units.

**Word help:** negligible = small enough to leave out of this model

**Sentence starter:** The … in my model represents … . Its label shows … .

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 07 Exit check

**Label** a thermometer reading and an energy-transfer arrow on the model.
**Explain** one difference between temperature and heat, including the usual units for each.

**Sentence starter:** Both … . They differ in … because … .

Initial T / phase: _____ Final T / phase: _____




Add an energy arrow; label mass, c or L as needed.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A small metal spoon is hotter than a large bath. Can temperature alone tell which contains more internal energy?
**Explain** what else you would need.

**Sentence starter:** My claim is … . The diagram/data show …, so … .

Sample A



Sample B



Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3